VISH RAO

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EDUCATION

UNIVERSITY OF PENNSYLVANIA, PERELMAN SCHOOL OF MEDICINE **Philadelphia, PA**
MD/PhD in Bioengineering **Expected May 2032**
- AI for medicine and biology; computer science

TSINGHUA UNIVERSITY, SCHWARZMAN COLLEGE **Beijing, China**
Master of Global Affairs and Leadership **June 2024**
- One of 141 scholars in the 2023–24 Schwarzman Scholars cohort
- Led engineering for the LLM-GLOBE capstone comparing cultural values in eight US and Chinese AI systems; the benchmark has since been used by groups at Mila/McGill, UCL, and the University of Tokyo
- **GPA:** 3.89/4.00; **Awards:** Tsinghua Outstanding Graduate Student (1 of 3 in Schwarzman College), Schwarzman Outstanding Capstone

COLUMBIA UNIVERSITY, SCHOOL OF ENGINEERING AND APPLIED SCIENCE **New York, NY**
Bachelor of Science (BS) Biomedical Engineering **May 2023**
- **GPA:** 4.09/4.00; **Awards:** *Summa Cum Laude, Tau Beta Pi*
- **Classes:** Neural Networks and Deep Learning, Machine Learning, Deep Learning in Biomedical Imaging, Natural Language Processing, Biostatistics, Differential Equations, Quantitative Physiology I/II, Physics Quantum Waves

AWARDS AND ACHIEVEMENTS

- First-author articles in *Nature*, *Nature Biotechnology*, and *Nature Biomedical Engineering*
- Author of 16 peer-reviewed publications (6 as first or co-first author) and 20 conference abstracts and presentations (500+ citations, h-index 9); co-reviewer for *Nature*; ad hoc reviewer for *The Laryngoscope*
- Schwarzman Scholarship recipient
- Rhodes Scholarship Finalist, Tau Beta Pi Fellowship recipient (<10% acceptance)
- John Kirkpatrick Young Investigator Award (Best Paper Award) from the *Society for Pediatric Radiology*
- 3x recipient of the MEEI-HMS Neuro-Laryngology Grant, 2x recipient of ISMRM Educational Stipend

RESEARCH EXPERIENCE

UNIVERSITY OF PENNSYLVANIA, PERELMAN SCHOOL OF MEDICINE **Philadelphia, PA**
PhD Student – Professor Zhi Huang (co-mentor: Professor Pranav Rajpurkar) August 2026 – Present
- Building a lab-in-the-loop AI scientist that uses virtual experiments with many AI models to choose the next real experiment; on public datasets, matched most answers from the complete data at about 5% of the cost
- Profiled cells on routine H&E slides with existing pathology models; predicted survival in 726 head and neck cancer patients about as well as CD3 immunohistochemistry
- Secured data from two randomized clinical trials (colorectal and kidney cancer) to evaluate models against patient outcomes

Rotation Student – Professor Katalin Susztak March 2025 – November 2025
- Developed a new foundation model (1B+ parameters) to jointly learn/predict histology and spatial single-cell omics across 20+ million cells
- Helped construct and benchmark NephroBase (3B+ parameters), a single-cell foundation model specialized for nephrology trained on over 40 million cells

HARVARD MEDICAL SCHOOL, DEPARTMENT OF BIOMEDICAL INFORMATICS **Boston, MA**
Summer Institute Scholar, then Researcher – Professor Pranav Rajpurkar May 2022 – Present
- Designed a novel method to generate a dataset of 1M+ annotated sentences for training and testing AI that detects errors in radiology reports; released as ReXErr-v1 on PhysioNet
- Led technical development of a deep-learning framework to verify correct endotracheal tube placement
- First-authored the *Nature* paper proposing the "AI resident" framework for medical image AI and a *Nature Biotechnology* paper on generalist biological AI, both with Professor Eric Topol (Scripps Research) as co-senior author

- Led the MAIDA consortium's neonatal study across 30 NICUs in 20 countries and territories (75 authors), adapting adult chest X-ray AI to newborns (*Nature Biomedical Engineering*, 2026); released the model openly
- Taught a session on natural language processing in the Medical AI Bootcamp (2022)

MASSACHUSETTS EYE AND EAR / HARVARD MEDICAL SCHOOL **Boston, MA**
Research Project Leader - Professors Phillip Song and Allen Feng March 2019 – September 2024
- Developed a new machine learning tool to screen for vocal disease using voice recordings
- Helped design and prospectively test a medical device to monitor and minimize force applied during surgery
- Constructed statistical models to analyze disease outcomes and treatment protocols from retrospective data (first author, *Head and Neck*, 2023)

COLUMBIA DEPARTMENT OF BIOMEDICAL ENGINEERING **New York, NY**
Research Project Leader - Professor Jia Guo January 2021 – September 2024
- Developed a transformer-based architecture for brain tissue segmentation that generalizes across datasets; applied to downstream tasks including schizophrenia classification and MRI contrast enhancement (first author, *Frontiers in Neuroimaging*, 2022; oral at ISMRM 2022)
- Devised new deep learning architecture to screen for schizophrenia using MRI scans
- Developed deep learning framework to artificially generate contrast-enhanced MRI from T1w structural scans

Leadership and Community Engagement

COLUMBIA DEPARTMENT OF BIOMEDICAL ENGINEERING **New York, NY**
Mentor – Professor Jia Guo's Lab **2021 – Present**
- Mentor students in the lab over calls, including on projects outside my own; currently co-mentoring undergraduates on a Drosophila connectome-inspired computer vision model (arXiv preprint, 2026)
- Project director for the Deep Learning in Biomedical Imaging course; helped mentor high school students in the BRAINYAC summer program

COLUMBIA RAAS **New York, NY**
Competition-Team Dancer **September 2022 – May 2023**
- Qualified for nationally competitive Garba-Raas Indian dance team at Columbia; performed several times on campus
- Accepted to and competed at national tournaments, including "Mania" at UIUC and "Sanedo" at UCSD

TAU BETA PI ENGINEERING HONOR SOCIETY **New York, NY**
Vice President and Treasurer for Columbia's Chapter **January 2022 – May 2023**
- Increased turnout by >75%; developed student incentives, increased advertisements across university channels, devised point system, hosted engaging events, and consistently polled student feedback

COLUMBIA ENGINEERING STUDENT COUNCIL **New York, NY**
Class President, Technology Representative, Transfer Student Representative **September 2020 – May 2023**
- Organized young alumni mentorship program; designed windowed scheduling system, aggregated common interests
- Advocated for policy initiatives to promote hybrid classes, zoom literacy, and pass/fail options during COVID

COLUMBIA ENGINEERING WITHOUT BORDERS **New York, NY**
Member, then Water Team Leader (2021 – 2022) **September 2020 – May 2023**
- Led the student team designing a water-system extension for 300+ residents of Amanfrom, Ghana; coordinated remotely with community members who built it

RED CROSS, LAHEY HOSPITAL, AND BEACON HOSPICE **Greater Boston, MA**
Volunteer **September 2020 – May 2023**
- Improved blood donation efficiency at the Red Cross by helping donors check in and supervising them post-donation
- At Lahey and Beacon: provided company to patients throughout their endoscopic and hospice care respectively; hosted events including a birthday party for a hospice patient

Publications and Presentations

** Equal contribution*

Peer Reviewed Publications:

1. **Rao VM***, Chen E*, Saenz A*, et al. Repurposing Adult AI Radiograph Models for Neonatal Care Through Continual Learning: An International Multicentre Study. *Nature Biomedical Engineering.* October 2026. DOI: 10.1038/s41551-026-01812-6.

2. Soylu S*, Okose OC*, Chen Y*, Wang B, Russell MD, Kamani D, **Rao VM**, et al. Prospective Pilot Study of a Smartphone App as a Screening Tool for Unilateral Vocal Fold Paresis/Paralysis. *European Archives of Oto-Rhino-Laryngology*. March 2026. DOI: 10.1007/s00405-025-09994-6.
3. **Rao VM**, Zhang S, Plosky BS, Hsu PD, Wang B, Zou J, Zitnik M, Topol EJ*, Rajpurkar P*. Generalist Biological Artificial Intelligence in Modeling the Language of Life. *Nature Biotechnology.* March 2026. DOI: 10.1038/s41587-026-03064-w.
4. **Rao VM**, Hla M, Moor M, Adithan S, Kwak S, Topol EJ*, Rajpurkar P*. Multimodal Generative AI for Medical Image Interpretation. *Nature*. 26 March 2025. DOI: 10.1038/s41586-025-08675-y.
5. **Rao VM***, Zhang S*, Acosta JN, Adithan S, Rajpurkar P. ReXErr: Synthesizing Clinically Meaningful Errors in Diagnostic Radiology Reports. *Pacific Symposium on Biocomputing*. January 2025. DOI: 10.1142/9789819807024_0006.
6. Low DM, **Rao VM**, Randolph G, Song PC*, Ghosh SS*. Identifying bias in models that detect vocal fold paralysis from audio recordings using explainable machine learning and clinician ratings. *PLOS Digit. Health.* 2024 May 30. DOI: 10.1371/journal.pdig.0000516.
7. Dermody SM, Kahng PW, **Rao VM**, et al. Color Match Following Free Flap Surgery in Head and Neck Reconstruction: A Colorimetric and Aesthetic Analysis. *PRS.* May 2024. DOI: 10.1097/PRS.0000000000010807.
8. Zhang J, **Rao VM**, et al. Detecting Schizophrenia with 3D Structural Brain MRI Using Deep Learning. *Sci. Rep*. 2023 September 2. DOI: 10.1038/s41598-023-41359-z.
9. **Rao VM**, Feng AL, Deschler DG, et al. Surgical Factors Affecting Regionally Metastatic Cutaneous Squamous Cell Carcinoma to the Parotid Gland. *Head and Neck*. 2023 Mar 8. DOI: 10.1002/hed.27304.
10. Wang TV, Feng AL, Puka E, **Rao VM**, et al. Reproducibility of Laryngeal Force Sensor Measurements. *Am. J. Otolaryngol*. March 2023. DOI: 10.1016/j.amjoto.2022.103782.
11. **Rao VM**, Wan Z, Arabshahi S, Ma DJ, Lee P-Y, Tian Y, Zhang X, Laine AF, Guo J. Improving Across-Dataset Brain Tissue Segmentation for MRI Imaging Using Transformer. *Frontiers in Neuroimaging.* 2022 Nov 21. DOI: 10.3389/fnimg.2022.1023481.
12. Rao GM, **Rao VM**, et al. Design, Development, and Face Validation of an Intubation Simulation Device Using Real-Time Force Data Feedback. *LIO*. 2022 Sep 15. DOI: 10.1002/lio2.916.
13. Wang AA, Feng AL, **Rao VM**, Naunheim MR, Juliano AF, Song PC. Clinical, Radiologic, and Endolaryngeal Findings in Laryngeal Fractures: A 15-Year Case Series. *OTO Open*. 2022 Feb 25. DOI: 10.1177/2473974X221080164.
14. Minaya NJ, **Rao VM**, Naunheim MR, Song PC. Laryngeal Subsite Analysis of Granulomatosis with Polyangiitis (Wegener's). *OTO Open*. 2021 July 15. DOI: 10.1177/2473974X211036394.
15. Feng AL, Puka E, Ciaramella A, **Rao VM**, et al. Laryngeal Force Sensor for Suspension Microlaryngoscopy: A Prospective Controlled Trial. *Otolaryngol Head Neck Surg*. 2021 Jan 5. DOI: 10.1177/0194599820982635.
16. Feng AL, Puka E, Ciaramella A, **Rao VM**, Sataloff RT, Naunheim MR, Song PC. Force Metrics and Suspension Times for Microlaryngoscopy Procedures. *J Voice*. 2019 Nov 29. DOI: 10.1016/j.jvoice.2019.10.017.

Preprints:

1. Li Z, Yamauchi A, Liu H, **Rao VM**, Guo J. From the Drosophila Visual Connectome to General-Purpose Computer Vision. *arXiv*. 6 Oct 2026. DOI: arXiv:2610.08418.
2. Li C, Ziyadeh E, Sharma Y, Dumoulin B, Levinsohn J, Ha E, Pan S, **Rao VM**, Subramaniyam M, Szegedy M, Zhang N, Susztak K. Nephrobase Cell+: Multimodal Single-Cell Foundation Model for Decoding Kidney Biology. *bioRxiv*. 30 Sep 2025. DOI: 10.1101/2025.09.30.679471.
3. Karinshak E*, Hu A*, Kong K*, **Rao VM***, Wang J*, Wang J, Zeng Y. LLM-globe: A benchmark evaluating the cultural values embedded in LLM output. *arXiv*. 9 Nov 2024. DOI: arXiv:2411.06032.
4. Cao A, **Rao VM**, Liu K, Liu X, Laine AF, Guo J. Tabsurfer: a hybrid deep learning architecture for subcortical segmentation. *arXiv*. 13 Dec 2023. DOI: arXiv:2312.08267.

Databases:

1. Rao VM, Zhang S, Acosta J, Adithan S, Rajpurkar P. ReXErr-v1: Clinically Meaningful Chest X-Ray Report Errors Derived from MIMIC-CXR (version 1.0.0). 2025 Mar 19. *PhysioNet*. RRID:SCR_007345. DOI: 10.13026/9dns-vd94.

Oral Presentations:

1. **Rao VM**, et al. ReXErr: Synthesizing Clinically Meaningful Errors in Diagnostic Radiology Reports. *Pacific Symposium on Biocomputing*. 2025. Hawaii, USA.
2. Stevenson LA, **Rao VM**, Feng AL, Song PC. Sex-Based Differences in Force Metrics During Suspension Microlaryngoscopy. *AAO-HNSF*. 2023. Nashville, TN.
3. **Rao VM**, et al. Improving Across-Dataset Brain Tissue Segmentation Using Transformer. *ISMRM.* 2022. London, UK.
4. Low DM, Randolph G, **Rao VM**, Ghosh SS*, Song PC*. Uncovering the important acoustic features for detecting vocal fold paralysis with explainable machine learning. *MEE Machine Learning Symposium.* 2020. Boston, MA
5. Feng AL, Puka E, Ciaramella A, Wang TV, **Rao VM**, Naunheim MR, Song PC. The Laryngeal Force Sensor Reduces Intraoperative Forces and Postoperative Complications: A Prospective Controlled Trial. *AAO-HNSF.* 2020. Boston, MA.

Poster Presentations:

1. Li C, Ziyadeh EM, **Rao VM**, Szegedy M, Susztak K. Nephrobase: Multimodal Single-Cell Foundation Model for Decoding Kidney Biology. *ASN Kidney Week*. 2025. J Am Soc Nephrol 36(10S): SA-PO0003.

2. **Rao VM**, et al. 3D Artificial Cerebral Blood Volume Generation from T1W Structural MRI. *ISMRM*. 2023. Toronto, Canada.
3. Tian Y, Zhang J, **Rao VM**, et al. Improving Across-Dataset Schizophrenia Classification with Structural Brain MRI Using Multi-Scale Transformer. *ISMRM*. 2023. Toronto, Canada.
4. Xin Z, **Rao VM**, et al. Synthesizing CT Image from Single-echo UTE-MRI using Multi-Task Framework Neural Network. *ISMRM*. 2023. Toronto, Canada.
5. Low DM, Randolph G, **Rao VM**, et al. Uncovering the important acoustic features for detecting vocal fold paralysis with explainable machine learning. *ALA.* 2023. Boston, MA.
6. Rao GM, Kim M, **Rao VM**, et al. Differences in force metrics between novices and experts during intubation simulation. *ABEA*. 2023. Boston, MA.
7. **Rao VM** et al. Pitfalls and Promises of Patient Reported Outcome Measures. *ALA*. 2023. Boston, MA.
8. Zhang J, Acosta N, **Rao VM**, Lee P, Gjerswold-Selleck S, Wan Z, Feng X, Liberman JA, Small SA, Guo J. Detecting schizophrenia using structural and synthetic cerebral blood volume MRI with deep learning. *ISMRM.* 2022. London, UK.
9. Tian Y, Zhang J, Zhu X, Lee P, **Rao VM**, Xin Z, Guo J. Classification of Schizophrenia with Structural MRI Using 3D-MIGR Transformer. *ISMRM.* 2022. London, UK.
10. **Rao VM**, Zhang J, Guo J. Exploring Hybrid CNN-Transformer for Schizophrenia Classification using Structural MRI. *ISMRM.* 2022. London, UK.
11. Rao GM, **Rao VM**, Feng AL, Juang J, Benoit J, Song PC. Design, Development, and Face Validation of an Intubation Simulation Training Using Real-Time Force Data Feedback. *ABEA*. 2022. Dallas, TX.
12. Wang TV, Feng AL, Puka E, **Rao VM**, Song PC. Reproducibility of Laryngeal Force Sensor Measurements. *AAO-HNSF*. 2021. Los Angeles, CA.
13. Wang AA, Feng AL, **Rao VM**, et al. Clinical, Radiologic, and Endolaryngeal Findings in Laryngeal Fractures. *AAO-HNSF*. 2021. Los Angeles, CA.
14. **Rao VM**, Feng AL, Deschler DG, et al. Surgical Factors Affecting Regionally Metastatic Cutaneous Squamous Cell Carcinoma to the Parotid Gland. *AHNS*. 2021. Chicago, IL
15. Minaya NJ, **Rao VM**, Song PC. Presenting Voice and Airway Manifestations in Granulomatosis with Polyangiitis. *AAO-HNSF*. 2020. Boston, MA.